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FINANCIAL CRIME SOLUTIONS

Financial crime solutions specialist who builds – not just operates – **Machine Learning**, **AI**, and **LLM**-powered systems. Engineers the optimum solution at minimum cost: free-tier and open-source tooling first, paid infrastructure only once value is proven. Deep in **Agentic AI**, **LLM** tooling, and **BigQuery ML**, with **SQL**, **Python**, **R**, **Power BI**, **Tableau**, **AWS**, **GCP**, **Oracle SQL**, and **NetReveal/RSA-Outseer** across banking, FinTech/crypto, and investment banking (**Bendigo & Adelaide Bank**, **Remitano**, **National Citizen Bank**). Proof in the build: self-directed AI products **Typology Extractor** and **PheChat**, shipped lean.

KEY SKILLS

AI & Machine Learning:
(Python, LLMs, NLP, Agentic AI, BigQuery ML)

Statistics & Detection Systems:
(SQL, Python, R, statistical calibration)

Automation:
(Oracle SQL, ETL, RPA-style pipelines)

Dashboard Building:
(Power BI, Metabase)

Leadership:
(Technology, Operations, Data, Group Risk)

- Built **Typology Extractor** (Project) — an Agentic AI/ETL pipeline transforming AUSTRAC guidance into structured data, supporting enforcement-gap detection & national scenario benchmarking
- Co-delivered **UAR-to-SMR AI Automation** (Pydantic-validated LLM prompting), cutting reporting time **80%**; built **PheChat** (Project), an Agentic RAG chatbot
- Certified: Claude Code, Claude API, Agent Skills & Subagents (Anthropic) and BigQuery ML Skill Badges (Google) (Certificates)
- **Scenario Management:** build, review & calibrate detection scenarios via optimisation algorithms & statistical analysis
- **Scenario Development Coordination (2025):** designed & calibrated **9** new detection scenarios, personally delivering **3** with a statistically significant true positive uplift
- **Annual Rule Review:** statistical testing & sampling cut false positives **5%**, removing **5,000+** alerts
- **Manual Alert Ingestion Pipeline:** Oracle SQL automation feeding **150+** alerts daily into NetReveal & RSA/Outseer with **99.9%** data integrity
- **UAR-to-SMR Automation:** data masking/rehydration, cutting reporting prep time **80%**
- Automated email marketing (**30%** CTR uplift); optimised ETL, cutting workload **25%**, cloud costs **30%**, runtime **70%**
- **Dashboard & Reporting:** Power BI dashboards & monthly reporting presented at **Board level**, eliminating **70%** of ad-hoc requests and cutting prioritisation time **60%**
- **Significant Dashboard Portfolio:** IFTI SWIFT network graph & BEN outlier detection dashboards spanning operational & detection uplift
- **Infrastructure Establishment:** stood up the team's Power BI Server & cloud workspace
- **Solution Delivery:** design & deliver financial crime detection solutions, partnering across Technology, Operations & Data teams
- **Scenario Development Coordination (2025):** personally delivered **3** of **9** scenarios while guiding others
- Supported the **Annual Rule Review** & led the **GitLab Migration**, improving version control & governance

PROJECTS

- **Typology Extractor** (typologyextractor.vercel.app) — Agentic AI and ETL pipeline that transforms AUSTRAC guidance documents into structured data for transaction monitoring framework comparison. Supports enforcement-

gap detection and national scenario benchmarking across financial crime typologies.

- **PheChat** (phechat.com) — Agentic RAG chatbot built for small and medium businesses, combining retrieval-augmented generation with autonomous agent workflows to answer customer queries and automate routine business tasks.

CERTIFICATES

- **A\ Introduction to Subagents** – Anthropic, Jul 2026
- **A\ Building with the Claude API** – Anthropic, Jul 2026
- **A\ Introduction to Agent Skills** – Anthropic, Jul 2026
- **A\ Claude Code in Action** – Anthropic, Jul 2026
- **A\ AI Fluency Framework & Foundations** – Anthropic, Jul 2026
-  **Engineer Data for Predictive Modeling with BigQuery ML** – Google Skill Badge, Jun 2025
-  **Create ML Models with BigQuery ML** – Google Skill Badge, Jun 2025
-  **Modernizing Data Lakes and Data Warehouses with GCP** – Coursera, Jun 2020

PROFESSIONAL EXPERIENCE

Analyst	Bendigo & Adelaide Bank	05/2021 - Present
<i>Intelligence and Analytics - Financial Crime Risk</i>		
<i>Australian banking group with 4 commercial banks and 1 digital bank under management</i>		
• Scenario Management: Build, review, and calibrate transaction monitoring detection scenarios, applying research and statistical analysis to drive the evergreen assessment of screening and detection systems. • Solution Delivery: Design and deliver financial crime detection solutions—including the AI-driven UAR-to-SMR automation pipeline and manual fraud/AML detection scenarios built outside vendor platforms—partnering across Technology, Operations, and Data teams. • Framework Set-Up: Set up and maintain the Power BI workspace and GitLab space underpinning the transaction monitoring product. • Dashboard Building: Own the Power BI dashboards as part of the transaction monitoring product—designing, developing, and maintaining them to provide key measures, KPIs, and guardrails.		

ACHIEVEMENTS

- **Annual Rule Review:** Executed and peer-reviewed the annual transaction monitoring rule review in **2022** and **2023**, supporting team members with statistical confidence testing, data sampling, and SQL-based scenario recalibration to cut false positives by **5%**, removing **5,000+** unnecessary alerts and saving **300+** investigation hours a year.
- **Scenario Development Coordination (2025):** Executed and supported team members in building **9** new detection scenarios in **2025**—personally delivering **3** with a statistically significant true positive uplift—while providing guidance on design and calibration.
- **UAR-to-SMR Automation (2025):** Co-delivered an end-to-end **AI-driven UAR-to-SMR submission automation** pipeline—implementing secure **data masking/rehydration** and **Pydantic-validated LLM prompting**—cutting regulatory reporting preparation time by **80%** while ensuring strict transparency and data privacy compliance.
- **Manual Alert Ingestion Pipeline:** Developed, implemented, and maintained a manual fraud/AML alert ingestion pipeline for detection scenarios built outside vendor platforms, feeding **150+ alerts daily** into **NetReveal** and **RSA/Outseer** with **99.9% data integrity**.
- **Rapid Incident Response:** Stood up a full detection service from scratch within **2–3 days** to support an urgent brand transition and incident response, maintaining uninterrupted transaction monitoring coverage under significant time pressure.
- **GitLab Migration:** Migrated manual detection scenarios from a shared network drive to **GitLab**, improving version control, auditability, and scenario governance.
- **Dashboard & Reporting:** Built Power BI dashboards and monthly reporting used by Operations and presented at **Board level**, eliminating **70%** of ad-hoc requests and cutting prioritisation time by **60%**.
- **Significant Dashboard Portfolio:** Delivered a suite of high-impact Power BI dashboards spanning **operational**

uplift (UAR, TM, and WLM case prioritisation) and **detection uplift**, including an **IFTI SWIFT network graph** for cross-border payment analysis and a **BEN outlier detection** dashboard for anomaly identification.

- **Infrastructure Establishment:** Initiated and established the team's **Power BI Server**, **Power BI cloud** workspace, and **GitLab** environment—foundational infrastructure enabling scenario version control and dashboard delivery.

Data Analyst

Remitano

03/2020 - 04/2021

International cryptocurrency exchange that matched \$500M+ in trading value and operated in more than 30 countries.

- Owned all data analytics requests, delivering dashboards and diagnostic reports in Python, SQL, and Metabase.
- Built marketing predictive models to automate and optimise strategy deployment.
- Designed, developed, and managed data crawlers, ETL pipelines, and Redshift database structures.
- Collaborated with engineering teams on complex SQL-based data transformations for scalable operational and marketing analytics.

ACHIEVEMENTS

- Automated email marketing via customer classification models (Python/SQL), boosting click-through rates by **30%** and enabling scalable campaign personalisation.
- Optimised ETL pipelines, indexing, and data normalisation in Redshift, slashing developer workload by **25%**, cloud costs by **30%**, and dashboard runtime by **70%**.
- Delivered **100% on-time** analytical reports and dashboards with consistent stakeholder satisfaction scores of **8/10 or higher**.
- Engineered complex SQL solutions (including recursive queries) for MLM commission calculations, processing **150K+** referral relationships across **4** levels to simplify payout computation, eliminate the need for a graph database, and power a **4-tier** referral program that drove **25%** of new user growth.

Investment Associate

National Citizen Bank

09/2014-02/2018

National Citizen Bank – A Commercial Bank with more than 20 years of operation

- Built financial models for property project estimation, company valuation, and debt restructuring, presenting findings to the Investment Council to support investment recommendations.
- Coordinated with in-house executives and clients on lending due diligence, gathering business information to support financial analysis and valuation.

ACHIEVEMENTS

- Secured Investment Council approval for **80%** of recommendations, generating a public equity portfolio **CAGR of 10%**.
- Cut investment approval time by **72 hours** and eliminated process conflicts by drafting a formal investment procedure covering accounting journal entries, authority levels, and document revision.

HONOURS & AWARDS

- Third place & Honourable Mention - Project SafeAus - Making Australia safer, Team VAIT - GovHack Australia, 09/2021
- #13 in public test and #32 in private test - PetFinder Competition, Team Vicohub-1 – Kaggle, 02/2019-03/2019
- First prize & Runner-up - Programming Competition – Department of Education and Training – 2007, 2008 & 2009

EDUCATION

- Master of Business Analytics – Deakin University, 2019
- Bachelor of General Business Management – National Economics University, 2013

VOLUNTEER ACTIVITIES & HOBBIES

- Tutored 100+ students in SQL and Python
- Won 2 Open Gov hackathon prizes
- Served as Coordinator of CFA Vietnam volunteer team
- Hobbies: Sports (Football/Soccer, Powerlifting), Building data-driven solutions for micro-businesses

REFERENCES

Available upon request